ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

Contact

www.linkedin.com/in/harini-v-380153441
(LinkedIn)

github.com/harini-vijayakumar
(GitHub)

Top Skills

C

Python (Programming Language)

Git & GitHub

Problem Solving

HARINI V

First-Year CSE Student (Batch 2026–2030) | Aspiring Software Developer | C & Python

Chennai, Tamil Nadu, India

Summary

" Strong fundamentals, steady curiosity and a habit of practising every day are what turn learning into good software. "

I am a first-year Computer Science and Engineering (CSE) student at St. Joseph's College of Engineering, Chennai. I have a basic programming foundation and am growing it through C and Python, while exploring software development, logical thinking and ways technology can solve everyday problems. My focus right now is to write cleaner code and turn what I learn into small working projects.

Outside the classroom, I have taken part in hackathons and technical activities, which taught me to work in a team, think creatively and stay calm while solving problems. I am also working on my confidence in communication and presenting my ideas. I am a quick learner who enjoys picking up new tools and is open to new challenges. My GitHub and LinkedIn profiles are growing alongside my studies.

Key Areas of Interest:

- **Software Development:** writing clean, structured programs and learning how applications are built.
- **Problem Solving:** practising logic and algorithmic thinking to break problems into clear steps.
- **Technology-Driven Solutions:** applying code to create practical, meaningful impact.
- **Open-Source & Version Control:** growing a public portfolio with Git and GitHub.

Tech Stack & Focus Tools:

- **Languages:** C, Python (Core Syntax, Data Structuring)
- **Core Tools:** Git, GitHub, VS Code
- **Strengths:** Quick learner, teamwork, creativity, adaptability

Experience

St. Joseph's College Of Engineering
Student
August 2026 - Present
Chennai

First-year undergraduate student pursuing a Bachelor of Engineering (B.E.) in Computer Science and Engineering. Currently building programming fundamentals in C and Python while developing structured coding habits and problem-solving skills.

Hackathons & Technical Activities

Participant

Took part in hackathons and technical activities, building creativity, teamwork and a problem-solving mindset.

Education

St. Joseph's College Of Engineering

COMPUTER SCIENCE AND ENGINEERING · (2026)

St. Joseph's College Of Engineering

Bachelor of Engineering, Computer Science and Engineering · (August 2026 - May 2030)

Career Goals

Short term: Strengthen programming fundamentals in C and Python, build projects, grow a strong GitHub portfolio and improve my communication confidence.

Long term: Become a confident software professional who keeps learning and builds meaningful technology solutions that create real-world impact.

Anand Prakash

Ethical Hacker • Cybersecurity Researcher • Entrepreneur — India
Profile compiled from public sources, October 2026

Biography

Anand Prakash is an Indian ethical hacker and entrepreneur from Bhadra, Rajasthan, now based in Bengaluru. He studied B.Tech Computer Science at VIT and began bug bounty hunting as a student. He became one of India's best-known bounty hunters (a top Facebook researcher, #3 on Twitter's programme), helped secure 400+ companies, and was named to Forbes 30 Under 30 Asia in 2017. He then turned that attacker's mindset into companies: AppSecure (2016) and PingSafe, acquired by SentinelOne in 2024 for a reported $100M+.

Career Timeline

2014	Security engineer at Flipkart.
2016	Founds AppSecure (red teaming). Facebook account-takeover flaw earns $15,000.
2017-19	Uber free-rides flaw (2017); Tinder flaw, about Rs. 4 lakh (2018); Uber account-takeover flaw, $6,500 (2019).
2020-22	Co-founds PingSafe with Nishant Mittal (year varies by source); $3.3M seed led by Peak XV Partners.
2023	Reported LinkedIn security flaw.
Jan 2024	SentinelOne acquires PingSafe; it becomes Singularity Cloud Native Security.

Career Goals (from his public talks and interviews)

- Make security attacker-driven: find real, exploitable weaknesses rather than theoretical ones.
- Secure cloud environments at scale, and promote India's role in global bug bounty research and startups.

Profiles

LinkedIn: linkedin.com/in/anandprakash6 (verified, shared by him on SquareX podcast)

GitHub: Not verified: no official profile found, and many unrelated people share the name. Check his LinkedIn for a link.

Projects & Schemes

PingSafe - cloud security (CNAPP) platform with an offensive security engine; customers included Flipkart and Razorpay. AppSecure - red teaming and security consulting. Bug bounty programmes - Facebook, Twitter, Uber, Tinder, LinkedIn and others. No government scheme linked to him was found.

Sources: Wikipedia, Global Indian, TechCrunch/Business Standard, freeCodeCamp, Nullcon 2022, SquareX and Techzine podcasts. Details such as his current role may have changed; confirm on LinkedIn.

NPTEL Summary

Summary of weekly lecture notes, Weeks 1–8

Week 1: Foundations of Soft Skills, Human Perception, and Self-Management

Week 1 establishes the foundational framework for soft skill enhancement and personal growth, highlighting that genuine soft skills development requires experiential learning and behavioral transformation rather than cosmetic rote learning. The curriculum examines how personal perceptions shape our interactions, noting that human relationships dictate eighty percent of our personal happiness. Stephen Covey's principle of "beginning with the end in mind" is introduced to urge learners to evaluate daily tasks against long-term objectives and eliminate non-value-adding habits.

Perception is analyzed through famous illustrations, showing that people often view identical realities through contradictory subjective lenses or prejudices. To overcome misperceptions, learners must seek first to understand before demanding to be understood, cultivating empathy to bridge interpersonal divides.

The curriculum transitions inward to self-management skills, establishing a core set of nine personal faculties: self-awareness, self-confidence, mindset, emotional balance, stress handling, coping with failure, patience/trustworthiness, perceptiveness, and spiritual intelligence. Living with both emotional balance and spiritual intelligence builds deep personal credibility, rendering external negative criticism inconsequential.

The pursuit of excellence is examined through psychological models, contrasting the "law of abundance" (which fosters optimism, courage, and fulfillment) with a "poverty of mind" (which breeds scarcity and fear-driven choices). Abraham Maslow's hierarchy is utilized to show that aspiring to self-actualization—the highest growth need—is vital, as individuals who leave their achievement potential unrealized remain unhappy. Finally, David McClelland's achievement motivation theory (N-Ach) and Danah Zohar's principles of spiritual quotient (SQ) demonstrate how individuals can view life as a meaningful project rooted in truth, goodness, and beauty.

Week 2: Conflict Resolution Paradigms and Stress Regulation

Week 2 tackles conflict resolution and stress management, framing both as inevitable realities of human interaction that must be embraced rather than avoided. Conflicts are defined as natural disagreements that can lead to constructive growth if handled properly, while destructive escalation typically stems from improper communication.

The progression from interpersonal negotiation to mediation and arbitration is explored, with the primary objective being the attainment of sustainable win-win solutions. Interpersonal case studies—such as husband-wife and father-son disputes—demonstrate the necessity of analyzing both viewpoints while remaining attentive to third-party contingencies that are often overlooked.

Becoming a conflict resolution expert requires shifting focus from demanding rigid solutions to asserting core needs. The lecture categorizes conflict into intrapersonal, interpersonal, and intergroup dimensions, identifying intrapersonal conflicts as the most difficult to resolve. When viewed constructively, conflict strengthens relationships, fuels self-growth, and boosts productivity thought styles such as collaboration or compromise. The focus then shifts to regulating stress, defined as an imbalance between situational

demand and individual supply. The syllabus differentiates between positive stress (eustress), which mobilizes energy, and distress, which degrades performance. Because employers view chronically stressed individuals as liabilities rather than resources, managing stress through physical fitness, adequate sleep, mindful planning, and simple relief rituals (such as walking in nature, deep breathing, or listening to music) is presented as an essential professional requirement.

Week 3: Habit Formation, the Habit Cycle, and Productivity

Week 3 explores the psychology of habits, defining them as unconscious behavioral responses triggered by environmental stimuli. Debunking the myth that habits can be formed or altered effortlessly, the instruction emphasizes that constructive habits require rigorous cultivation, whereas destructive habits form easily through peer influence and leave lasting marks on character.

The habit cycle is broken down into its fundamental mechanics: thought/stimulus leading to action/reaction, which receives reward or punishment, solidifies into habit, builds character, and ultimately yields life success or failure. Drawing from literary examples such as R.K. Narayan’s The Guide and William Golding’s Lord of the Flies, the lectures illustrate that while the capacity for base impulses exists within us, the moral resolve to change and foster noble habits also resides entirely within our conscious choice.

Breaking destructive habits requires understanding the neurological role of dopamine, which induces feelings of pleasure and can either reward constructive challenges or anchor harmful addictions. A cornerstone of the week's study is the Zeigarnik effect, which describes the human brain's natural anxiety and cognitive burden when confronted with unfinished tasks. Left incomplete, tasks consume mental bandwidth and create internal dissonance. However, by understanding that procrastination is merely an initial starting hurdle, learners can harness the Zeigarnik effect constructively: once an activity is started, the intrinsic drive for completion propels the individual through to the end.

Finally, the week reviews common habits of highly successful people, highlighting the value of diligent hard work, completing extra tasks willingly, working without constant clock-watching, maintaining intense focus on primary goals, and replacing bad routines with positive habits.

Week 4: Listening Dynamics and Professional Telephone Skills

Week 4 focuses on foundational communication skills, identifying active listening and telephone etiquette as critical competencies. Active listening is presented as an energetic, purposeful mental process distinct from passive physiological hearing.

Effective listening requires an active seven-stage sequence: reception, selection, organization, assimilation, interpretation, evaluation, and response. Learners are warned against common misconceptions, such as believing that only speaking expends energy or that good speakers do not need to listen.

Overcoming physical noise, physiological discomforts, and psychological barriers (such as prejudice, partial attention, or emotional bias toward a speaker) requires whole-body engagement, empathetic attentiveness, and the strategic use of silence.

The second half of the week details telephone communication, an arena where the absence of visual non-verbal feedback increases the risk of misunderstanding. Basic telephone skills include preparing pen and paper before answering, smiling while speaking to convey warmth tonally, pacing speech, and

providing verbal encouragements like "I see" and "go on". Advanced practices focus on clear enunciation to eliminate verbal confusion between similar-sounding words, desk organization, and adherence to professional call frameworks when delivering difficult news. Essential skills for handling demanding or irate callers involve voice modulation, politeness, jargon avoidance, and proactive problem anticipation. Mastering these telephone behaviors transforms phone interactions from potential career pitfalls into valuable opportunities to project competence and earn respect.

Week 5: Technology, Mobile Etiquette, and Digital Netiquette

Week 5 evaluates the profound impact of modern digital technology on human personality and interpersonal relationships. Drawing on Donna Haraway's concept of the "cyborg," the lecture explores how the blurring boundary between the biological and the mechanical causes individuals to treat machines with human-like care while treating people like cold machines. Critical questions of control, benefit, and choice are raised, alongside Albert Einstein's warning that technology might eventually surpass human interaction. The proliferation of mobile devices has turned humans into "moborgs" who succumb to nomophobia (mobile addiction) and obsessive-compulsive checking, displacing immediate relationships in favor of virtual connections. To counter this, learners are urged to establish device-free boundaries, treat phones as utilitarian tools rather than masters, and practice mobile courtesy by silencing devices in formal meetings and avoiding texts when face-to-face communication is possible.

The curriculum also establishes professional written standards for digital correspondence, organizing them around the five cardinal principles of soft skills: planning, preparedness, persuasiveness, presentability, and perseverance. Demonstrating the real-world fallout of unprofessional emails, the course highlights the hazards of missing salutations, all-caps text, colloquial slang, blended languages, and neglected grammar. The week concludes with comprehensive guidelines for internet etiquette, or "netiquette". Because digital messages can be stored permanently and remain under the recipient's control, communicators must remember that there is a human being behind the screen, choose words with care, express emotional nuance through thoughtful language or emoticons, and uphold strict ethical standards.

Week 6: Communication Architecture, Barriers, and NonVerbal Assessments

Week 6 investigates the mechanics of effective communication and the internal and organizational factors that disrupt it. Effective communication is defined as the deliberate ability to generate an intended response. The communication process is modeled through Harold Lasswell's formula—who sends what to whom, through which channel, and with what effect. A message's success depends on clarity, conciseness, conviction, timing, and genuine empathy.

Communication breakdowns are traced to psychological filters within both sender and receiver, including mismatched frames of reference, encoding/decoding errors, and emotional interference. At the organizational level, communication flows downward, horizontally, and upward through written, oral, and electronic channels. However, long reporting chains, administrative hierarchies, unnecessary transfer points, and a lack of workplace trust can severely distort information. This distortion is illustrated through real-world examples of downward messaging errors and product development misunderstandings. To address these challenges, communicators must adapt their choice of medium, remove fears surrounding change, and verify mutual understanding.

The week concludes by shifting attention toward non-verbal communication, administering pre-thinking assessments to uncover students' baseline knowledge and dispel widespread misconceptions regarding body language.

Week 7: Non-Verbal Dynamics, Universal Cues, and Evaluative Settings

Week 7 expands into non-verbal communication, noting that up to ninety-three percent of communication effectiveness is governed by body language. Because non-verbal cues stem from subconscious feelings and are difficult to fake, they serve as reliable indicators of authentic emotion. Non-verbal behavior fulfills five key functions: repeating, complementing, contradicting, regulating, or substituting for verbal statements. The lectures cover the core domains of non-verbal analysis: kinesics (body movement and gestures), facial expressions, oculesics (eye gaze), haptics (touch), proxemics (interpersonal space), chronemics (time), and paralinguistics (vocal tone and pauses).

Learners are guided through the interpretation of non-verbal cues across three primary dimensions: immediacy (liking), arousal (responsiveness), and dominance (power balance). Accurate interpretation requires applying the "3 Cs"—analyzing context, reading gesture clusters, and detecting baseline behavioral changes—while accounting for external factors like culture and temperature to avoid premature judgments.

The focus then shifts to evaluative scenarios like job interviews and group discussions. For interviews, candidates are coached on punctuality, dress codes, firm handshakes, consistent eye contact, and composed posture. In group discussions, participants are cautioned against aggressive posturing, nervous fidgeting (such as leg shaking or nail-biting), or withdrawing into silence. Instead, they are taught to project calm confidence through open-palm gestures, pleasant facial expressions, and attentive nodding.

Week 8: Presentation Mastery, Reading Efficiency, and Human Relations

Week 8 covers public speaking, reading comprehension, and professional ethics. Public speaking anxiety is traced to its root causes: fear of humiliation, fear of the unfamiliar, and dread of the unexpected. Presenters can overcome this stage fright by mastering their material, maintaining a confident self-image, channeling adrenaline into enthusiasm, and visualizing a standing ovation.

Professional delivery requires viewing the venue beforehand, welcoming attendees, articulating three or four main points clearly, and balancing informative content with engaging anecdotes. Visual aids must be designed thoughtfully: presenters should use concise bullet points and clear fonts, avoid blocking the screen, and ensure spare copies are ready. Presenters must also maintain open postures, project their voices clearly, and avoid hiding behind podiums.

The final section explores effective reading techniques, explaining how reading broadens mental perspective and strengthens vocabulary. Core techniques include previewing/sampling to select relevant material, skimming for main themes, scanning for specific data, clustering words to improve reading speed, and close reading to analyze nuance in philosophical or literary works.

The course concludes by reinforcing that professional competence and soft skills are meaningless without foundational trust and ethical integrity. Governed by the golden rule—treating others as one wishes to be treated—learners are reminded that circumstances do not create character, but merely reveal it. Because circumstances do not create character, but merely reveal it.

SWOC Analysis

My Software Engineering Roadmap to 2030

Harini V
First-Year B.E. Computer Science and Engineering
St. Joseph's College of Engineering, Chennai

Strengths - Weaknesses - Opportunities - Challenges

Profile & Career Vision

Where I am today and where I want to go

1 Who I am

First-year CSE student with basic programming skills, a quick learner who enjoys working in a team.

2 Vision

Become a confident software professional who builds technology solutions with real-world impact.

3 Objective

Turn my programming basics into real projects, stronger problem solving and confident communication.

4 Framework

Use the SWOC matrix to find my gaps and trends, then convert them into a year-by-year plan.

SWOC Matrix at a Glance

Internal factors on top, external factors below

Strengths — Internal

- Basic programming skills
- Quick learner
- Good at team work

Weaknesses — Internal

- Nervous communicating in front of others
- Lack of confidence in speaking
- Overthinking unfamiliar tasks

Opportunities — External 2030

- Hackathons and competitions
- Online learning resources
- College labs and technical events

Challenges — External 2030

- Strong competition
- Balancing academics with technical learning
- Fast-changing technology

SWOC Analysis | Harini V 3

Strengths

What I already bring to the table

1 Basic programming skills

I have a basic foundation in programming that I can build on step by step.

2 Quick learner

I pick up new concepts fast, which matters most in a field that changes quickly.

3 Good at team work

I work well with others, which helps in hackathons, labs and group projects.

4 Willingness to improve

I am open to feedback and keen to keep growing every semester.

SWOC Analysis | Harini V 4

Weaknesses

Honest gaps I am working on

1 Nervous while communicating

I feel nervous when speaking in front of others, especially in groups.

2 Lack of confidence in speaking

I hold back my ideas even when I know the answer.

3 Overthinking unfamiliar tasks

I spend too long thinking before starting something new instead of trying it.

4 Limited hands-on experience

I need more projects and practice to move from concepts to application.

Opportunities by 2030

Where I can grow as a software engineer

1 Hackathons

Learn by building and working with a team under real deadlines.

2 Competitions

Coding and technical contests help me test and sharpen my skills.

3 Online learning resources

Free courses, tutorials and communities make skill-building affordable.

4 College labs

Lab sessions give regular hands-on practice with real tools.

5 Technical events

Workshops, seminars and symposiums build knowledge and a network.

6 Growing tech fields

AI, cloud, cybersecurity and data all need strong programmers by 2030.

Challenges by 2030

What I must prepare for

1 Strong competition

Many engineering students aim for the same roles, so projects and skills matter beyond marks.

2 Balancing academics and tech

Managing coursework along with extra technical learning needs good time planning.

3 Fast-changing technology

Tools and languages shift quickly, so continuous learning is a must.

4 Coding automation

Simple tasks may be done by AI, so I must stand out through problem solving.

5 Industry skill gap

Classroom learning alone may not match what companies expect.

6 Consistency over time

Staying disciplined and motivated across four years of study.

SWOC Analysis | Harini V 7

Turning SWOC into Strategy

Pairing my internal factors with external trends

Strength + Opportunity

Use my quick learning and team work in hackathons and competitions to build real projects.

Strength + Challenge

Use my quick learning to keep pace with new tools and stand out in a competitive field.

Weakness + Opportunity

Use events, labs and team activities to practise speaking and build confidence step by step.

Weakness + Challenge

Start tasks early instead of overthinking, and plan my week so academics and tech learning both fit.

SWOC Analysis | Harini V 8

Action Plan: Closing the Gaps

Concrete habits to turn weaknesses into strengths

1. **Speak up regularly**
   Answer or ask one question in every class and present in small groups first.
2. **Practise presenting**
   Explain my projects aloud to friends, then give short talks at college events.
3. **Act before overthinking**
   Give myself 10 minutes to plan, then start the unfamiliar task and improve as I go.
4. **Build small projects**
   Finish one small project each month and keep a clean GitHub with a README.

Action Plan: Using the Opportunities

Making the most of what 2030 offers

1. **Join hackathons**
   Participate in hackathons with a team to learn by building.
2. **Enter competitions**
   Try coding contests and technical competitions each semester.
3. **Learn online**
   Follow a free online course alongside college studies, a little every week.
4. **Use labs and events**
   Make full use of college labs and attend technical events and workshops.

4-Year Milestone Roadmap

A suggested path from basics to industry-ready (2026 - 2030)

Year	Primary Focus	Target Deliverable	Skill Goal
Year 1 2026-27	Programming basics, problem solving, communication	3 small projects + 1 hackathon	Strong programming basics
Year 2 2027-28	Data structures, algorithms, web basics	Full-stack mini project	Internship-ready skills
Year 3 2028-29	Specialization (AI, cloud or embedded)	Capstone-style project + competition entry	Industry-level depth
Year 4 2029-30	Interview prep, advanced projects	Portfolio + final-year project	Placement or higher studies

Conclusion & Key Takeaways

Where this analysis leaves me

1 Start strong

My programming basics, quick learning and team work are a solid launch point.

2 Grow steadily

Building speaking confidence and hands-on projects is my priority.

3 Aim for 2030

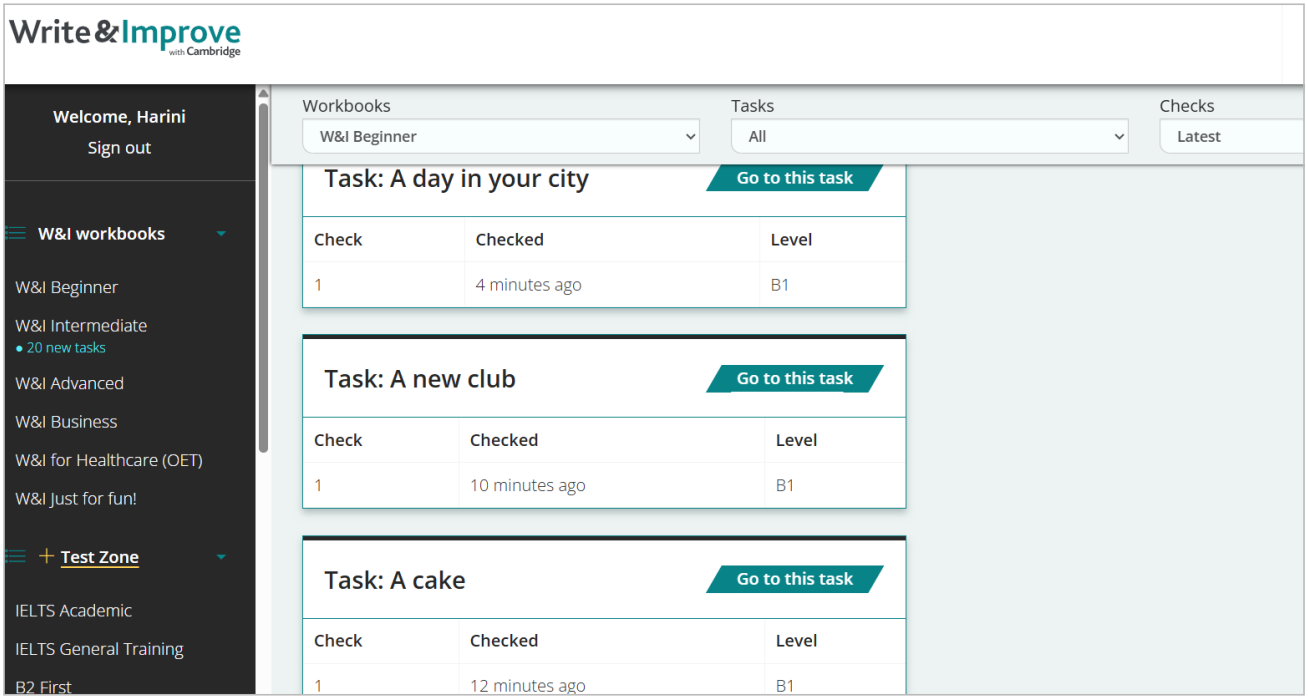
With steady effort I can ride trends in AI, cloud and software to build technology with real-world impact.

Strong basics today, real-world impact by 2030.

Harini V | CSE, St. Joseph's College of Engineering

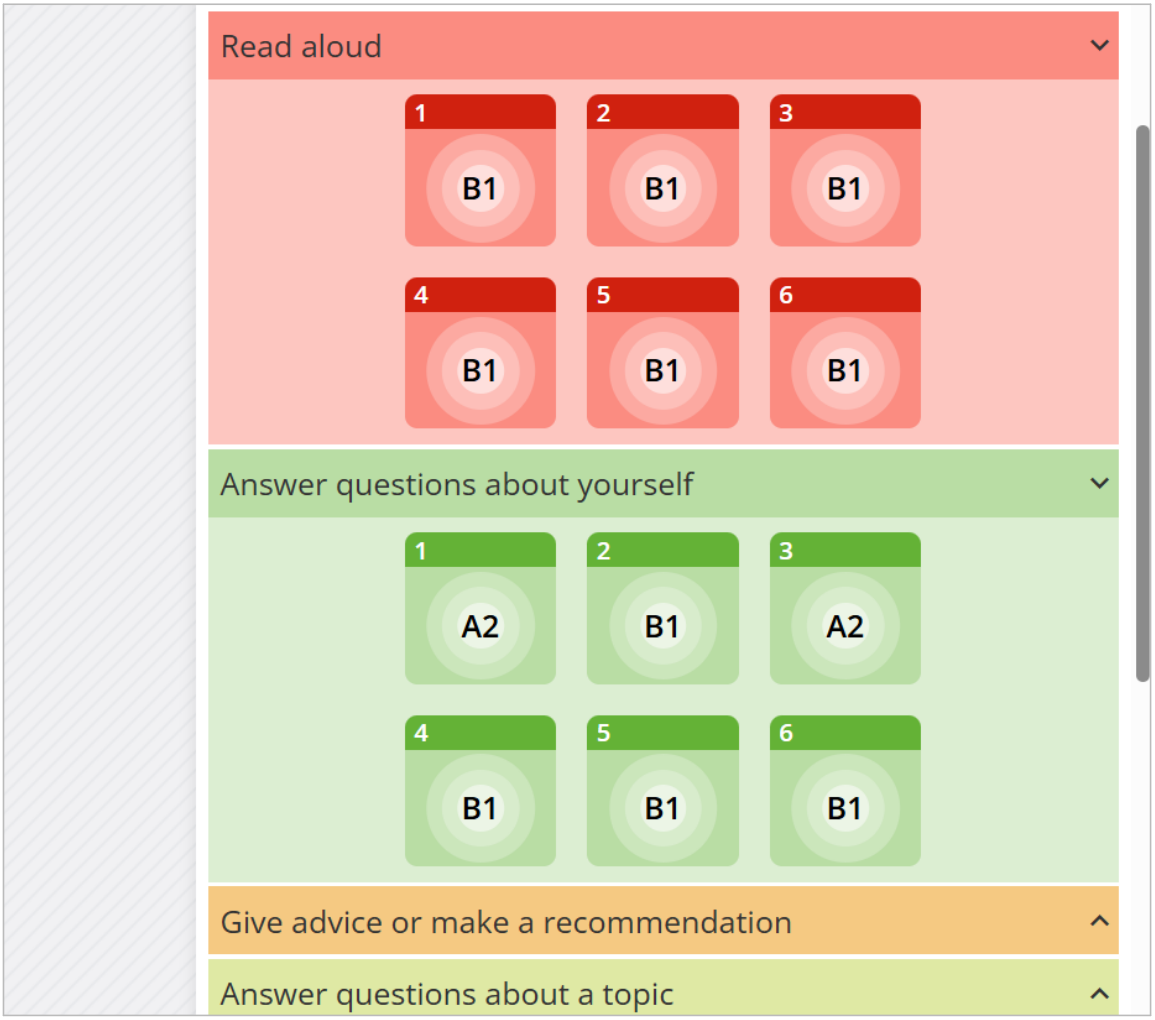
Write and Improve

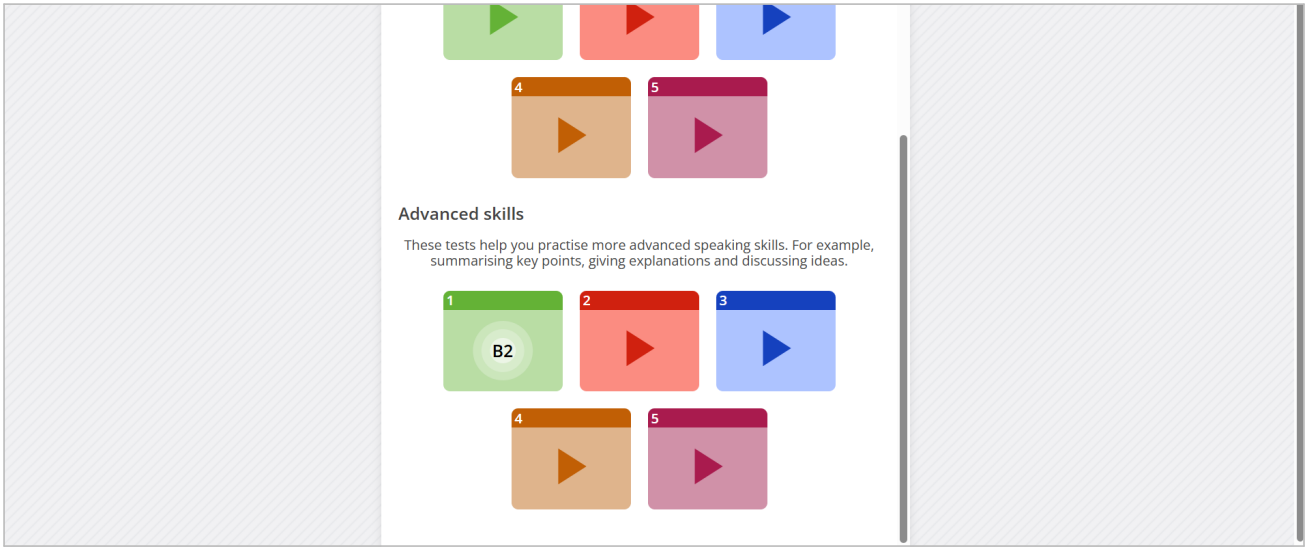
Write & Improve (Cambridge): W&I; Beginner tasks and checked levels



Speak and Improve

Speak & Improve: practice tests and levels





Grammar and Proficiency Check

Reviewing writing for coherence and proficiency level

Reviewing writing for coherence and proficiency level.

Overall result: B2 (lower B2)

This is a short text, so the level is an estimate. I also need to correct my earlier answer. I said there were no passive voice or participle structures, but there are, and they are what push your level up to B2.

Grammar check

I found **no grammatical errors**. Tenses, articles, punctuation, and reported speech are all correct. Structures worth noting:

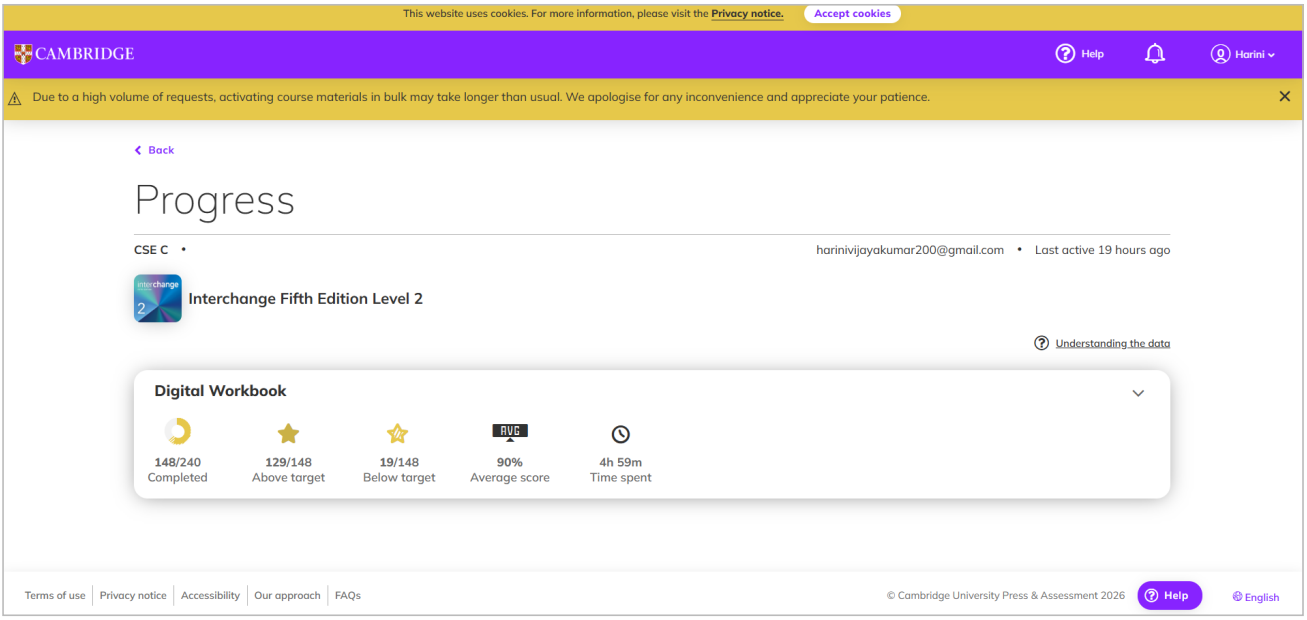
- Passive voice with a relative clause: *"programming languages, which are translated into a form..."*
- Participle phrase: *"binary numbers, consisting of zeros and ones"*
- Defining relative clause without a pronoun: *"a form the computer can process"*
- Past continuous interrupted by past simple: *"was working... when she asked"*
- Gerund after a preposition: *"After listening to his explanation"*

Style issues (not errors):

- "explained" appears three times, plus "also explained", which is repetitive.
- "her brother" is repeated in nearly every s nce. Pronouns or "he" would flow better.

Digital Workbook Progress

Cambridge One: Interchange Fifth Edition Level 2, digital workbook progress summary



Worksheet 1 – Part C

5 × 13 = 65 Marks

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for every account.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.
5. Check that a website address begins with “https” and shows a padlock before entering any sensitive information.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or the National Cyber Crime Helpline (1930).

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Selected Question: 27.a

Bharatanatyam: The Dance That Carries Tamil Nadu’s Soul

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student’s formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

Question 28 (b) Formal letter of request to the Principal

Selected Question: 28.b

Harini V
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

1 November 2026

To
The Principal
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission.

Thank you for your kind consideration.

Yours obediently,

Harini V
(Register No.: [Register No.])

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a

AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

Question 30 (b) Personal email: Orientation programme

Selected Question: 30.b

To: [friend's email address]
Subject: My first-year orientation programme at college

Dear Ramya,

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Harini V

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.

3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

BusBuddy: Project Ideation

Smart College Bus Tracking System

1. Project Objective

To design a simple system that helps students check a college bus's location and estimated arrival time, so they can plan when to leave for their bus stop and reduce unnecessary waiting.

2. Problem Statement

Students often do not know where their college bus is or when it will reach their stop. This uncertainty can lead to long waiting times, missed buses, and repeated calls to drivers or friends. A simple, accessible bus-tracking system could make this information easier to obtain.

3. Proposed Solution

BusBuddy is a web or mobile application that displays the bus's latest shared GPS location on a map. Students select their route or stop to view an estimated arrival time. The system can also show route details and provide an alert when the bus is approaching. For a student prototype, a phone's GPS can be used to share the bus location; a live deployment would require permission and coordination with the college and driver.

4. Key Features

- Live bus location on a map (when GPS sharing and internet access are available).
- Route and bus-stop information.
- Estimated time of arrival (ETA) for selected stops.
- Optional approaching-bus notification.
- Simple interface for students and a basic driver/admin view to share or manage location.

5. What Makes It Unique?

BusBuddy focuses on the daily needs of college students rather than general city transport. It can combine the college's own routes, familiar stop names, and simple arrival alerts in one lightweight app. A useful first version can be built with a small number of routes instead of an expensive transport-management system.

6. How It Works

1. The driver or authorized device enables location sharing for the bus.
2. The app sends the GPS coordinates to a server or database over the internet.
3. The student opens BusBuddy and selects a bus route or stop.
4. The app shows the latest bus position and estimates arrival time using location and route distance.
5. If enabled, the app alerts the student when the bus is near the selected stop.

7. Technology Stack

Frontend: HTML, CSS, JavaScript (or Flutter for a mobile app)

Map: OpenStreetMap with Leaflet

Backend: Python with Flask

Database: Firebase or SQLite for a prototype

Location: Smartphone GPS and internet connection

8. Expected Impact

- Reduce uncertainty and unnecessary waiting at bus stops.
- Help students plan their departure more effectively.
- Make route and arrival information easier to access.
- Improve communication between students and college transport staff.

9. Feasibility

The project is feasible at an intermediate student level. A prototype can start with one route, a few stops, a map, and a manually shared or phone-based GPS location. Open-source mapping tools and common web technologies keep the cost low. Testing can be done with simulated bus movement before using real location data.

10. Limitations

- Location and ETA accuracy depend on GPS quality, internet connectivity, and traffic conditions.
- Live tracking requires permission, a location-sharing device, and cooperation from the college or driver.
- Continuous GPS use can consume phone battery and mobile data.
- Student location privacy and access permissions must be handled carefully.
- A prototype’s ETA may be approximate and may not reflect sudden route changes.

Project scope note: For an assignment prototype, demonstrate one route with sample stops and simulated or consent-based GPS data. Do not claim live tracking or accurate ETAs until these functions have been tested.

LinkedIn and GitHub Profiles

Part A: LinkedIn profile

Harini V

B.E. CSE Student | Aspiring Software Engineer | C, Python & Web Development | AI & Technology Enthusiast

St. Joseph's College Of Engineering

Chennai, Tamil Nadu, India

Part B: GitHub profile README

GitHub username: harini-vijayakumar

About Me

Hi! I'm an engineering student passionate about technology, programming, and cybersecurity.

- Currently learning programming and computer science fundamentals.
- Exploring C programming and problem-solving.
- Aspiring Cybersecurity Analyst.
- Interested in learning new technologies and building practical projects.
- Always curious to learn, improve, and grow in the tech world.

My 2026 Learning Goals

- Strengthen my C programming fundamentals.
- Learn Python programming.
- Explore Linux and command-line basics.
- Understand computer networks and network security.
- Learn cybersecurity fundamentals and ethical hacking concepts.
- Build beginner-friendly projects and upload them to GitHub.
- Practice problem-solving and improve my technical skills.

Areas I'm Interested In

- Cybersecurity
- Ethical Hacking
- Network Security
- Linux & Operating Systems
- Programming & Software Development
- Vulnerability Assessment
- Problem-Solving

Tools & Technologies I'm Currently Working On

- Programming: C
- Learning Next: Python
- Version Control: Git & GitHub
- Exploring: Linux
- Cybersecurity: Security Fundamentals & Networking Basics
- Other Skills: Algorithms, Flowcharts & Pseudocode

My Mindset

“Learn consistently. Build practically. Think securely. Grow continuously.”

Thanks for visiting my profile! Feel free to explore my projects and follow my learning journey.